

MANNAN SHUKLA

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TECHNICAL SKILLS

Kubernetes (operators, controllers, CRDs), Docker, Helm, ArgoCD, Terraform, Cilium • Go, Python, Bash, TypeScript • vLLM, continuous batching, PagedAttention/KV cache, CUDA, Slurm, DCGM, NCCL • PyTorch, Hugging Face, LoRA • AWS (EC2, ECS, RDS, S3, DynamoDB, IAM) • Prometheus, Grafana • GitOps, CI/CD, GitHub Actions • Auth0, JWT, RBAC

EXPERIENCE

Software Engineer

July 2025 - Present

Voltage Park / Lightning AI

San Francisco, CA & Redmond, WA

- Shipped hot model-weight reloading for **vLLM** deployments via a **Go** sidecar, swapping fine-tuned checkpoints in-place in seconds vs. 10–30 min GPU redeloys
- Designed an async/batch inference platform: a durable **Postgres**-backed task queue (SKIP LOCKED claims, leases) with pull-based sidecar workers and queue-depth autoscaling (scale-to-zero)
- Built core pieces of an **S3**-compatible object store (**VAST**-backed): a health-aware **P2C** client-side load balancer over CNode IPs with lock-free in-flight tracking and automatic failover, plus **Terraform/Vault**-managed **RDS** infra
- Built distributed inference orchestration on **Kubernetes** in **Go**: deployment-target controllers and **DynamoDB** event sourcing to coordinate fleets of **GPU** serving replicas
- Built and operate a multi-tenant managed **Kubernetes** platform across **GitOps** (ArgoCD), **Terraform/Helm** IaC, **Cilium/Envoy** networking, **NVIDIA GPU Operator**, and **kube-prometheus-stack** observability, with tenant-isolation hardening
- Built a **Slurm** GPU health-check suite for distributed training (ECC, PCIe AER, **NCCL** stalls, **DCGM** checks, thermal) that auto-drains unhealthy nodes to protect multi-node jobs
- On-call for production **H100** clusters; debugged distributed networking/scheduling across **WireGuard**, **CoreDNS**, **Cilium**, **MPI Operator**, and **GPU Operator**

Undergraduate Researcher: SuperAgent

Apr 2024 - Apr 2025

Rutgers University

New Brunswick, NJ

- Built an **LLM** agent recommendation system hitting 83% accuracy vs. **GPT-4** with a 1B model via **LoRA** fine-tuning (**Unsloth**)

Partnerships + Business Development

Feb 2025 - May 2025

Nas.io

New York, NY

- Drove influencer partnerships and client onboarding for a Series A startup, generating \$2K GMV in one month while 50x'ing outreach via a **Python/Klaviyo** automation pipeline

PROJECTS

BYOA: Bring Your Own Agent

May 2026

- Built a **Kubernetes** platform that spins up agents on any **HuggingFace** model in one click, leveraging **NVIDIA Dynamo** for disaggregated prefill/decode serving, KV-cache-aware routing, and GPU autoscaling behind OpenAI-compatible endpoints

tochi.one: AI property management

June 2025

- Built payment, lease-writing, and communication agents for landlords using **Agno**, with **Twilio** for communication agents and the **Stripe** API for payments

AdBlockIRL: AR ad-blocking proof-of-concept for Meta Quest

May 2025

- Trained **YOLOv8** on **Roboflow** to detect 100+ concurrent ads at 5ms latency, applying real-time mosaic censoring via **OpenCV**; won \$2000 at Bain Capital Ventures AI Hackathon

SlopGen: Automated short-form video generator

June 2024 - Sep 2024

- Built end-to-end pipeline using **yt-dlp**, **WhisperX**, and **llama3** via **ollama** achieving 92% segmentation accuracy (15x faster than manual), with custom TTS voices via **edge-tts**

EDUCATION

Rutgers University - New Brunswick

Bachelor of Science - Computer Science • Cum Laude

- Linux User Group (President), Road to Silicon Valley Program